Sovereignty on the Spectrum

The Case Against Admitting Starlink into Indian Cyberspace

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The outer frontier of Indian national security no longer runs along the Line of Actual Control alone. It now runs five hundred and fifty kilometres above it, through the orbital shells of a privately held American constellation whose on and off switch is wielded, in public and without apology, by a single foreign citizen. The question of whether to admit SpaceX Starlink into Indian cyberspace is therefore not a question of broadband policy. It is a question of whether the Republic retains, within a domain increasingly ungoverned by terrestrial writ, the final authority to communicate, to intercept, to interdict, and to be let alone.

I. The Geopolitical Instrument

Starlink is not a telecommunications provider in the ordinary sense. It is a low Earth orbit constellation that by the first quarter of 2026 crossed ten thousand active satellites, operated by SpaceX, the private vehicle of Mr. Elon Musk.[1] The same corporate entity contracts with the United States Department of Defense and the National Reconnaissance Office for Starshield, a hardened national security line of business constructed around a classified contract of approximately 1.8 billion United States dollars first disclosed by The Wall Street Journal in February 2024 and detailed by Reuters the following month.[2] The commercial and the military services are formally distinguished by the operator; United States Air Force filings and reporting by FedScoop have nonetheless established that they share launch cadence, inter satellite laser architecture and, in certain government deployments, user terminals that attach to both constellations.[3] They are, in material terms, cousins on the same orbital infrastructure.

Mr. Musk has established, through conduct rather than theory, that he regards the beam of his constellation as a personal foreign policy instrument. In September 2022 he withheld Starlink coverage over Russian held Crimea in order to defeat a Ukrainian naval drone strike against the Black Sea Fleet at Sevastopol, a decision first disclosed in Walter Isaacson's authorised biography, subsequently refined by Isaacson's own clarification that the service was never activated over the peninsula, and ultimately not disputed by SpaceX or by Mr. Musk himself.[4] On 14 June 2025, during the twelve day war between Israel and the Islamic Republic of Iran, he confirmed on his platform X that "the beams are on" over Iranian territory.[5] One week later, on 21 June 2025, Mr. Richard Grenell, United States Presidential Envoy for Special Missions, publicly requested on the same platform that Starlink be made free for Iranians.[6] On 13 January 2026, following a telephone call between Mr. Donald Trump and Mr. Musk, SpaceX waived subscription fees for all terminals operating within Iran, a fact confirmed by Bloomberg and CNN.[7] These are not commercial decisions. They are sovereign acts carried out by a private party. To permit such an instrument to blanket Indian territory without ironclad local custody of its ground segment is to delegate a slice of national communications policy to a jurisdiction over which the Republic exercises no writ.

II. A Strategic Reading

The Starlink question is best examined through three frameworks drawn from the formal literature on strategic interaction: the Stackelberg leader and follower model, the problem of credible commitment formalised by

Thomas Schelling in the context of deterrence, and the Stackelberg security game studied in the computer science literature on adversarial systems.[8]

In the Stackelberg formulation the player who moves first and whose move is publicly observed enjoys structural advantage, because the follower optimises against a fixed constraint. The Republic of India, as the sovereign licensor, is the natural first mover. It sets the conditions of entry; the operator chooses whether to comply. This advantage, however, is a perishable asset. Once the operator attains installed base, that is, once millions of terminals are embedded in Indian households, enterprises, hospitals and government installations, the roles invert. The operator becomes the effective first mover in any subsequent dispute because the withdrawal of service imposes immediate welfare losses on Indian users that no government can politically absorb. The sovereign leverage must therefore be exhausted at the point of entry, which is the present moment, and not conserved for a future negotiation that will not exist on symmetric terms.

The Schelling framework compounds this logic. Credible commitment, in Schelling's analysis, requires the irrevocable removal of one's own options so that reversal becomes impossible or prohibitively costly. A state can bind a foreign operator only through mechanisms the operator cannot subsequently evade: physical gateways under Indian custody and jurisdiction, data residency enforced by cryptographic audit rather than by assurance, and lawful interception architecture whose keys reside with Indian agencies. Corporate codes of conduct, bespoke architectures announced at trade conferences, and executive statements of intent do not meet the Schelling test. They are reversible by the operator at any moment, and the Iran precedent demonstrates that they are reversed when geopolitical imperatives in Washington override commercial undertakings elsewhere.

The third framework, originating in the work of Tambe and others on security games, instructs that the defender's optimal strategy is not the one that fares best against a cooperative counterpart but the one whose worst case outcome against a fully adversarial counterpart remains tolerable.[9] Applied to the present matter, the licensing regime must be engineered on the stipulation that at some point in the multi decade life of the constellation the operator will act contrary to Indian interests, whether under coercion by its home state, through unilateral commercial decision, or by compromise at the hands of a third party. The licence conditions must survive that eventuality without requiring the counterpart's cooperation.

A further structural difficulty is information asymmetry. The operator possesses private information concerning satellite firmware, inter satellite laser link routing, terminal provisioning logic and gateway failover architecture that the regulator cannot independently verify at scale. In the vocabulary of mechanism design, the Republic must therefore structure the contract such that incentive compatible disclosure by the operator coincides with sovereign oversight. The minimal incentive compatible mechanism is the one toward which the Ministry of Home Affairs has gravitated: no commercial launch is permitted until an Indian authority can, at any moment and without foreign cooperation, sever the network's reach into Indian territory.

The folk theorem of infinitely repeated games is sometimes invoked to suggest that long term commercial self interest will discipline the operator through the shadow of future transactions. This reasoning fails in the present case. The relevant discount factor for the operator is set by the geopolitical return on compliance in Washington, not by the commercial return on compliance in Delhi, and the two diverge sharply under conditions of crisis. The operator has already selected, in Ukraine and in Iran, actions consistent with the

former and against the explicit will of host governments. There is no analytic basis to presume that the pattern would not recur over the Indian subcontinent.

III. The Terrestrial Case: Low Cost Internet and the India Stack

The argument that affordable satellite connectivity constitutes a public good must be weighed against the Indian experience of the past decade, which demonstrates that terrestrial networks deliver scale faster, cheaper and under sovereign jurisdiction. The entry of Reliance Jio in September 2016 and the ensuing collapse of mobile data tariffs rendered India, by several international measures, the lowest cost mobile data market in the world. That price collapse created the economic foundation for the Unified Payments Interface, which in August 2026 processed 24.51 billion transactions valued at 29.82 lakh crore rupees, a volume the International Monetary Fund has recognised as the largest real time payment system in the world.[10]

The same terrestrial fabric carries Aadhaar authentication, DigiLocker, FASTag, CoWIN, the Direct Benefit Transfer pipeline, the Open Network for Digital Commerce, the Goods and Services Tax invoicing system, and real time income tax filing. This composite architecture, frequently termed Digital Public Infrastructure or the India Stack, has been deliberately engineered as a sovereign stack: open specifications, domestic hosting, Indian regulatory oversight. If a foreign satellite operator with gateways outside India were to become a mainstream last mile pipe, each of these flows would transit a route the Republic neither owns nor fully monitors. The sovereign stack would then rest, in material part, upon a foreign backbone. The marginal utility of satellite coverage is genuine only for the residual pockets of unconnected territory that terrestrial networks cannot economically reach, and such pockets can be served through indigenous constellations, through ventures operating under the Indian National Space Promotion and Authorisation Centre, and through supervised partnerships with BSNL and private satcom licensees.

IV. The Iranian Precedent

In June 2025, during the twelve day war between Israel and the Islamic Republic of Iran, Mr. Musk publicly confirmed that Starlink beams were switched on over Iranian territory. By that date, industry analysts estimated between twenty thousand and fifty thousand terminals were in circulation inside the country, having arrived through smuggling networks and, following the January 2026 crackdown on protests, through approximately six thousand terminals covertly shipped by the United States government and reported by The Wall Street Journal.[11] The sequence corresponds to a well documented pattern in the literature on information operations: hardware is pre positioned under the cover of commercial trade, the beam is activated during a period of political instability, and the resulting flow is framed in liberatory terms.

The Iranian response was conducted through electronic warfare. Mobile jamming units were deployed against Starlink's Ku band uplink in the 10.9 to 14 gigahertz range and against the Ka band in the 18 to 40 gigahertz range, the two radio frequency windows upon which the constellation's user and gateway links depend. Of greater operational consequence was the targeting of Global Positioning System signals, upon which Starlink terminals rely for geolocation and satellite handoff: spoofing broadcast counterfeit positioning signals that caused terminals to lose synchronisation. Open source analysis has attributed the Iranian capability to the indigenous Cobra V8 electronic warfare system, augmented by the Russian Kalinka system transferred following the twelve day war.[12] Independent reporting cited by Al Jazeera, Forbes and the German defence

publication ESUT placed degradation of Starlink links inside Iran during January 2026 at between thirty and eighty per cent, concentrated in the regions of greatest protest activity.[13] The instructive conclusion for every observer government, India included, is that a foreign satellite communications network does not remain a neutral utility when geopolitical conditions shift. It becomes a lever. The neutralisation of such a lever requires sophisticated and costly electronic warfare capabilities that few states possess, and that India has no principled reason to maintain against a service operating over its own sovereign territory.

V. The Domestic Preview: Manipur and the Andamans

The threat has already presented itself within Indian jurisdiction. On 13 December 2024 the Spear Corps of the Indian Army, operating in insurgency affected Manipur in a joint operation with Manipur Police and Assam Rifles, recovered a Starlink satellite antenna and router from a cache seized at Keirao Khunou in Imphal East district. The device bore the markings RPF and PLA, denoting the Revolutionary People's Front and its armed wing, the People's Liberation Army of Manipur, both proscribed under Indian law.[14] Alongside the Starlink equipment were recovered an MA4 assault rifle used by the Myanmar Army, a pistol, grenades and twenty metres of FTP cable. Investigators concluded that the hardware had been smuggled across the porous Myanmar border, where the Myanmar Internet Project has estimated approximately three thousand Starlink terminals in operational use with ethnic resistance organisations, a figure that the National Unity Government of Myanmar places at over four thousand.[15] In late November 2024, police in the Andaman and Nicobar Islands recovered a Starlink Mini from a Myanmarese fishing trawler apprehended near Barren Island, carrying 6,016 kilograms of methamphetamine the Andaman and Nicobar Police valued at retail at 360 billion rupees, or approximately 4.25 billion United States dollars.[16]

Mr. Musk responded to the Manipur disclosure on 17 December 2024 with a public assertion on X that "Starlink satellite beams are turned off over India". The salient technical point, however, is that the hardware does not require authorised service to constitute a security problem. It requires only that the beam be activated, by any party, at any point in the future. The sovereign decision to withhold service is at present a matter of corporate discretion, not of enforceable Indian legal control. Internet shutdowns, controversial in their own right, remain a lawful instrument available to the Indian state under the Temporary Suspension of Telecom Services (Public Emergency or Public Safety) Rules, 2017. A satellite communications link that bypasses terrestrial infrastructure renders those shutdowns inoperative. In a theatre of active armed conflict, this is not a civil liberty. It is a tactical advantage transferred to armed non state actors.

VI. The Cybersecurity Architecture

Three technical considerations render an unregulated Starlink entry particularly hazardous for the Republic. The first concerns gateway control. In its standard architecture, Starlink routes user traffic through the ground station geographically or topologically nearest to the terminal, which may be located outside Indian jurisdiction. Lawful interception under section 69 of the Information Technology Act, 2000 and the equivalent provisions of the Telecommunications Act, 2023 can be exercised only where the gateway resides on Indian soil and under Indian jurisdiction. In its absence, Indian investigative agencies cannot serve warrants in operable form, cannot inspect traffic metadata, and cannot interdict unlawful content in real time.

The second consideration is terminal fingerprinting. Each dish is cryptographically bound to an account provisioned by SpaceX. The operator possesses, through global positioning system lock and inter satellite link telemetry, metre scale knowledge of the location of every active terminal at every moment. This constitutes a continuously updated register of users, necessarily including any Indian official, enterprise, defence installation or sensitive site connected to the network. No foreign operator should hold such a register for Indian territory without strict data residency, independent audit by an Indian authority, and legally enforceable Indian access to the operational logs.

The third consideration is attack surface. A satellite ground segment is a priority target for state sponsored intrusion. On 24 February 2022, approximately one hour before Russian ground forces entered Ukraine, the KA SAT satellite communications network operated by Viasat suffered a cyberattack that disabled tens of thousands of modems across central and eastern Europe through the AcidRain data wiping malware. On 10 May 2022, the United States government, the United Kingdom's National Cyber Security Centre and the Council of the European Union jointly attributed the operation to the Russian military intelligence service, the GRU.[17] Should an analogous intrusion compromise SpaceX operational networks, the compromise would cascade into every connected Indian customer. The national power grid, financial infrastructure, and defence installations would thereby acquire a new transit path over which the Indian Computer Emergency Response Team has no visibility. This constitutes an inadmissible blind spot in the risk posture of a state that faces persistent cyber probing from at least two neighbouring state actors.

VII. The Regulatory Line and Its Preservation

The Indian framework has crystallised. Starlink must obtain security clearance from the Ministry of Home Affairs in addition to its Global Mobile Personal Communication by Satellite licence from the Department of Telecommunications, awarded on 6 June 2025, and its authorisation from the Indian National Space Promotion and Authorisation Centre, awarded on 8 July 2025 and valid until July 2030.[18] All domestic traffic must flow through Indian gateways. A monitoring and control centre must be established within India. Data of Indian subscribers must remain in India. Each gateway and hub site requires its own clearance. On 5 May 2025, the Department of Telecommunications issued twenty nine supplementary security conditions governing data localisation, website blocking, metadata collection and integration of the indigenous NavIC positioning system; satellite to satellite traffic was expressly barred from circumventing Indian gateways.[19] In June 2026 the draft Telecommunications (Spectrum Assignment by Administrative Process) Rules, published by the Department of Telecommunications, required explicit governmental permission for the interconnection of any satellite communications network with public telecommunications networks and conditioned commercial rollout on security clearance even after spectrum allocation.[20] These are not protectionist measures. They constitute the minimum conditions under which any sovereign state may claim meaningful control over communications flowing across its territory.

At the India Mobile Congress on 7 October 2026, Ms. Lauren Dreyer, Vice President for Starlink Business Operations at SpaceX, stated that the company had built its operations "specifically for India in recognition of Indian security requirements and other regulations, establishing controls at every layer" and would keep Indian user data within India; she disclosed that twenty gateway sites with hundreds of antennas had already been constructed in the country.[21] The appropriate posture of the Republic is to verify, not to accept assurance. The final security clearance should be withheld until each technical, legal and operational commitment has been

independently audited by an Indian agency competent to do so, until a demonstrated sovereign kill switch resides unambiguously with the Indian state, and until the terms are enforceable against SpaceX in Indian courts without recourse to foreign arbitration.

VIII. Conclusion

The question before the Republic is not whether India requires further connectivity. It does. The question is whether that connectivity shall traverse a conduit whose on and off switch is held in California, whose terminals have already been recovered in the possession of insurgents in Manipur and smugglers in the Andaman Sea, and whose principal has demonstrated, in Ukraine and in Iran, that he will deploy its coverage as an instrument of foreign policy. The Ministry of Home Affairs has read the strategic terrain correctly. The clearance must remain withheld until the question of sovereignty is answered in paper, in silicon, and in enforceable Indian law. The line must be held.

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